Microcontroller functionalities

1. Assume that an input from a sensor is designated as Low when it is actively in operation and floating otherwise. Your task is to determine if the sensor is in operation or not. Describe, with the aid of a diagram, how you would connect the sensor's input to the microcontroller to prevent it from entering a floating state.
   A floating input may cause random or unstable readings in the microcontroller.
   Solution: Use a Pull-up Resistor
   When the sensor is inactive (floating), the pull-up resistor pulls the MCU input **HIGH (logic 1)**.
   When the sensor is active (drives LOW), it connects the input to **GND**, reading **LOW (logic 0)**.
   Thus, the input never "floats" between undefined voltage levels.
   Some MCUs allow **internal pull-up resistors** via software configuration.

2. Write a simple pseudo-code or C++ program that generates a delay using a loop-based approach. Explain the role of a Timer/Counter in generating more accurate delays compared to such software-based delays.

```
void delay_loop(unsigned long t)
{
 for (unsigned long i = 0; i < t; i++)
   // Empty loop to waste time
 }
```

   A Timer/Counter is a hardware module that: Counts clock pulses independently of the CPU & Can generate precise time intervals by comparing counts. Hence, hardware timers are far more accurate and efficient than loop-based delays.

3. In a line-following and obstacle-avoiding robot, both sensors may trigger actions simultaneously. Explain how interrupts can be used to handle both events efficiently without missing sensor inputs.
   Use **interrupts** so both sensor inputs are handled immediately. When either sensor changes state, it triggers its interrupt service routine (ISR). ISRs execute independently of the main loop, ensuring no event is missed. It enables simultaneous event handling without polling or missed inputs.

4. Explain the function of an Analog-to-Digital Converter (ADC) in microcontrollers. Assume a 10-bit ADC with a reference voltage of 5V and an analog input of 2.75V. Calculate the corresponding digital output value.
   An **Analog-to-Digital Converter (ADC)** transforms a continuous analog voltage into a discrete digital value that the MCU can process.
   ADC resolution = 10 bits → 2^{10} = 1024 levels

$$\text{Digital Output} = \frac{V_{in}}{V_{ref}} \times (2^{10} - 1)$$

$$= \frac{2.75}{5.00} \times 1023 = 562.65 \approx \boxed{563}$$

5. Suppose that a GPS module is connected to a microcontroller using a UART interface. List the parameters that needs to be matched between the microcontroller and the GPS module in order to have a successful communication between the two.

Baud rate	Data transmission speed
Data bits	8/ 16/ 32 bits
Parity bit	None / Even / Odd
Stop bits	1 or 2 bits
Voltage levels	Logic levels must be compatible (3.3 V or 5 V)

If any parameter mismatches → data corruption or communication failure.

6. Differentiate SPI and I2C communication protocols considering their architecture, speed, and duplex nature. Illustrate using figures when necessary

Feature	**SPI (Serial Peripheral Interface)**	**I²C (Inter-Integrated Circuit)**
Lines Used	4 lines → MISO, MOSI, SCK, SS	2 lines → SDA, SCL
Architecture	Full-duplex Master–Slave	Half-duplex Master–Slave
Speed	Up to several MHz	Up to 400 kHz (standard)
Addressing	Slaves selected by separate SS lines	Each device has a unique address
Complexity	More wires, simple protocol	Fewer wires, complex protocol